



Check if the Sentence Is Pangram

Company: Goldman Sachs

Year: Not specified

Difficulty: Easy

Topic: Strings, HashSet, Array

Source: LeetCode 1832 – Check if the Sentence Is Pangram

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Problem Description

A **pangram** is a sentence that contains **every letter of the English alphabet at least once**.

You are given a string sentence containing only lowercase English letters.

Your task is to check whether all 26 letters from:

a to z

appear at least once in the sentence.

Return:

true

if the sentence is a pangram, otherwise return:

false

Input Format

- The first line contains a string sentence consisting of lowercase English letters.

Output Format

- Print true if sentence contains every letter from a to z at least once.
- Otherwise, print false.

Constraints

- $1 \leq \text{sentence.length} \leq 1000$
- sentence consists of lowercase English letters.

Example 1

Input:

```
sentence = "thequickbrownfoxjumpsoverthelazydog"
```

Output:

true

Explanation

The sentence contains every letter of the English alphabet at least once.

Therefore, it is a pangram.

Example 2

Input:

```
sentence = "leetcode"
```

Output:

false

Explanation

The string does not contain all 26 letters of the English alphabet.

Therefore, it is not a pangram.

Approach to Solve This Problem

The main idea is to keep track of which letters have appeared in the string.

Since there are exactly **26 lowercase English letters**, we can use a boolean array of size 26.

For every character:

'a' → index 0

'b' → index 1

'c' → index 2

...

'z' → index 25

We can find the position using:

index = character - 'a'

For example:

'a' - 'a' = 0

'c' - 'a' = 2

'z' - 'a' = 25

Whenever we see a character, mark its position as true.

After processing the entire string, check all 26 positions.

If every position is true, then every alphabet letter appeared at least once, so the sentence is a pangram.

Java Solution

```
import java.util.*;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
String sentence = sc.nextLine();

boolean[] seen = new boolean[26];

for (char ch : sentence.toCharArray()) {
    int index = ch - 'a';
    seen[index] = true;
}

for (int i = 0; i < 26; i++) {

    if (!seen[i]) {
        System.out.println(false);
        return;
    }
}

System.out.println(true);
}
```

Solution:

1. **Read the input:** main() reads the string sentence, matching the Input Format.
2. **Create a boolean array:** Create a boolean array seen of size 26. Each position represents one letter of the English alphabet.

3. **Mark each letter:** Traverse every character of sentence. Convert the character into an index using `ch - 'a'` and mark that position as true.
4. **Check all alphabet letters:** After processing the string, traverse the seen array from index 0 to 25. If any position is still false, that corresponding alphabet letter was not present in the sentence.
5. **Determine the result:** If every position in seen is true, then all 26 letters appeared and the sentence is a pangram. Otherwise, it is not a pangram.
6. **Print the result:** Print true if all 26 letters are present; otherwise, print false, matching the Output Format.

Complexity Analysis

- **Time Complexity:** $O(N)$ — we traverse the sentence once, followed by a constant 26 checks.
- **Space Complexity:** $O(1)$ — the boolean array always contains exactly 26 elements.